Large-Scale Hyperspectral Image Restoration via a Superpixel Distributed Algorithm Based on Graph Signal Processing

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***Abstract*— Hyperspectral image (HSI) is often disturbed by various kinds of noise, which brings great challenges to subsequent applications. Many of the existing restoration algorithms do not scale well for HSI with large size. This article proposes a novel mixed-noise removal method for HSI with large size, by leveraging the superpixel segmentation-based technology and distributed algorithm based on graph signal processing (GSP). First, the underlying structure of the HSI is modeled by a two-layer architecture graph. The upper layer, called skeleton graph, is a rough graph constructed using the modified k-nearest-neighborhood algorithm and its nodes correspond to a series of superpixels formed by HSI segmentation. The skeleton graph can efficiently characterize the intercorrelations between superpixels, while preserving the boundary information and reducing the computational complexity. The lower layer, called detailed graph consisting of a series of local graphs which are constructed to model the similarities between pixels. Second, based on the two-layer graph architecture, the HSI restoration problem is formulated as a series of optimization problems each of which resides on a subgraph. In each optimization problem, a graph Laplacian regularization (GLR) is defined and incorporated into a low-rank (LR)-based model. Third, a novel distributed algorithm is tailored for the restoration problem, using the information interaction between the nodes of skeleton graph and subgraphs. Numerical experiments conducted on both synthetic and real-world datasets demonstrate the effectiveness of the proposed restoration algorithm compared with existing methods.**

***Index Terms*— Distributed algorithm, graph Laplacian regularization (GLR), graph signal processing (GSP), large-scale hyperspectral image (HSI) denoising, superpixel segmentation.**

I. Introduction

HYPERSPECTRAL image (HSI) is composed of tens or even hundreds of gray-level images corresponding to spectral channels, which has high spectral resolution and abundant texture information. Based on these HSI characteristics, the combination of spatial images and spectral bands can provide favorable conditions for the extraction of object features. Thus, it is widely used in food quality and safety assessment [1], medical diagnosis and image-guided surgery [2], precision agriculture [3], and other fields [4]. However, HSI is often disturbed by various types of noise in the process of acquisition and transmission. These noises bring great challenges to the subsequent analysis, especially in feature extraction and target recognition, which will greatly reduce the reliability of the processing. Therefore, how to effectively remove the noise of HSI has become a crucial problem.

Currently, various denoising approaches have been proposed for the restoration of HSI. A traditional approach for HSI denoising is to describe each band as a gray-level image and then apply 2-D denoising methods to remove noise band by band, such as the collaborative filtering of groups of similar patches block-matching and 3D filtering (BM3D) [5] and the weighted nuclear norm minimization (WNNM) [6]. However, these methods ignore the high correlations among the spectral bands or spatial pixels, resulting in relatively poor restoration performance. In view of this point, many HSI denoising methods have been proposed to take these correlations into account. In [7], a hybrid spatial–spectral derivative-domain wavelet shrinkage model is proposed, which considers the dissimilarity of signal regularity in spatial and spectral dimensions. In [8] and [9], an image operator called total variation (TV) is applied for HSI restoration, which preserves the spatial piecewise smoothness (PWS) property of HSI. In [10], a spatial–spectral TV (SSTV) regularization is proposed to characterize the spatial and the spectral correlation of HSI, which is an expansion of TV operator. In [11] and [12], low-rank (LR) matrix approximation-based methods are proposed to exploit low-dimensional structures in high-dimensional data. Furthermore, tensor decomposition based methods are also proposed to exploit the LR property of HSI, such as the lower rank tensor approximation (LRTA) [13], the robust low-rank tensor recovery (RLRTR) [14], the tensor subspace low-rank learning with non-local prior (TSLRLN) [15], and the weighted low-rank tensor recovery (WLRTR) [16]. Moreover, some methods that combine different representation of LR property are proposed, such as the Kronecker-basis-representation based tensor sparsity measure (KBR) [17], the global local factorization (GLF) [18], the double low-rank (DLR) matrix decomposition [19], and the factor group sparsity-regularized nonconvex low-rank approximation (FGSLR) [20]. These LR-based methods effectively represent the strong spectral correlation of HSI. To characterize the LR and PWS property of HSI simultaneously, the methods proposed in [21], [22] and [23] combine LR-based theory and TV operator or its expansion.

Besides, nonlocal-based methods are proposed to characterize the nonlocal self-similarity of HSI, which can significantly

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improve the restoration effect by segmenting the image into overlapping square patches, clustering the similar patches into groups, and then processing the groups. Some of nonlocal-based methods, such as the nonlocal tensor-ring (NLTR) approximation [24] and the non-local meets global (NGmeet) restoration paradigm [25], restore the HSI group by group, which only consider the nonlocal self-similarity but ignore the PWS property in the group. The others consider nonlocal self-similarity as a regularization term in the model and combine the regularization term with other global regularization term, such as TV and SSTV. With the combined regularization terms, the methods in [26] and [27] achieve excellent restoration performance, but they can not simply restore the HSI group by group since it globally restores the entire image using a global optimization problem.

Different from the nonlocal-based methods, superpixel segmentation-based methods, such as the superpixel segmentation and low-rank representation (SSLRR) [28] and the superpixel segmentation of smooth band (SSSB) [29], group the pixels of HSI into a homogeneous region, whose shape is adaptive to boundary information and can be irregular. Thus, the superpixel segmentation-based methods can effectively preserve the boundary information, which is not shared by the nonlocal-based methods since they use the square patch-based strategy. However, the existing superpixel segmentation-based methods, such as SSLRR and SSSB, perform poor on HSI restoration, which is partially owing to that they have not considered the PWS property of HSI and the similarity between superpixels. Moreover, by simply incorporating the PWS property of HSI into superpixel segmentation-based methods, the performance is also restrictive due to the irregular domain data [30], [31]. In view of these points, we resort to the graph signal processing (GSP), a multidisciplinary research area, which has been recently developed to process irregular network data. It employs graph topology to describe the correlation between network data and comprehensively builds a theoretical framework for network data processing, which greatly expands the theoretical boundary of classical signal processing. Therefore, we are motivated to develop a new way to restore the HSI by leveraging GSP and superpixel segmentation-based technology.

Furthermore, deep learning-based methods, e.g., convolutional neural network (CNN) and deep neural network (DNN), are developed as powerful tools for HSI denoising in recent years, which have excellent nonlinear fitting and feature expression ability and achieve well performance [32]. In [33], a CNN-based method is proposed by modeling spatial and spectral features through dilated convolution and multichannel filters, respectively. In [34], a combined spatial-spectral deep residual CNN (HSID-CNN) is proposed, which is based on a residual CNN. Besides, there are various DNN-based methods proposed for HSI denoising domain, such as the feed-forward denoising CNN (DnCNN) [35], the separable 3D CNN (SP3DCNN) [36], the HSI single denoising CNN (HSI-SDeCNN) [37], and the method in [38], all of which have achieved good denoising performance. Although these deep learning-based methods are utilized as a powerful feature extraction tool, the "black-box" model feeds on massive paired training data simulating the network inputs (noisy images) and outputs (clean images), which lacks interpretability of mathematical modeling through noises and priors [39]. Moreover, the current deep learning-based methods are often designed for single Gaussian noise. These methods are relatively simple and are not applicable for more complex real noise situations, and require training models for different levels of noise leading to poor versatility [19].

In addition, the existing methods always focus on a small-scale HSI or a subregion of large-scale HSI and they exhibit poor scalability on large HSI since they employ the centralized operators in general. Besides, several restoration methods transform the 3-D HSI dataset into a 2-D matrix, an ill-posed matrix where the spatial dimension greatly exceeds the spectral one, resulting in blurring and a loss of details in denoised results [40]. The loss of boundary information and texture details will be more serious for large size HSI. Therefore, how to improve the scalability of the restoration methods is a key problem [41]. The distributed algorithm with GSP can be the candidate to resolve this problem and they have been successfully applied in graph signal reconstructions [42], [43], [44], [45]. These distributed methods with GSP always exchange information with neighbors during each iteration, which are potential to achieve better performance than the nonlocal or superpixel segmentation-based methods which restore each image patch without information exchange.

This article proposes an efficient restoration algorithm for large-scale HSI, called graph Laplacian regularization and superpixel segmentation-based distributed algorithm (GLRSSDA), which leverages the superpixel segmentation-based technology and distributed algorithm based on GSP. The main contributions of this article are summarized as follows.

1) A two-layer architecture graph is proposed to model the underlying structure of the HSI. The upper skeleton graph can well characterize the intercorrelations between superpixels, while preserving the boundary information and reducing the computational complexity. The lower detailed graph can well characterize the similarities between pixels.
2) Based on the intercorrelations between superpixels and topology of skeleton graph, the detailed graph is decomposed into a series of overlapping subgraphs capturing the inherent structural similarity, particularly the spatial correlation between pixels. Accordingly, the global restoration is proceeded on the subgraphs, i.e., we restore the entire HSI region by region. Then a GLR is defined to enhance the representation of the PWS property of each subgraph, which is incorporated into an LR-based model to formulate the restoration problem on each subgraph into an optimization problem.
3) Based on the restoration problem formulation on subgraphs, a novel distributed algorithm is tailored to restore HSI, which possesses good scalability.
4) Numerical experiments conducted on both synthetic and real-world datasets with large size confirm the effectiveness of the proposed algorithm.

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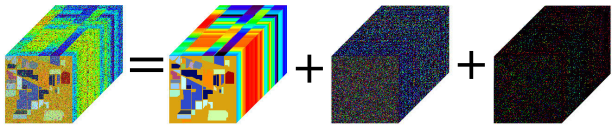


Fig. 1. Decomposition of the HSI model.

The rest of this article is organized as follows. We first introduce the problem formulation of HSI in Section II. In Section III, some preliminary knowledge is introduced. In Section IV, the proposed method is represented. Then, numerical experiments on both synthesis and real datasets are carried out to illustrate the effectiveness of the model in Section V. In Section VI, we conclude this article.

II. Problem Formulation

A. Notations

We use the common convention of representing matrices and vectors with bold letters and scalars with normal letters in the rest of this article. For a matrix $\mathbf{A}$, denote its rank, transpose, conjugate transpose, Frobenius norm, operator norm on ℓ^p, $1 \le p \le \infty$ and trace, by $\text{rank}(\mathbf{A})$, $\mathbf{A}^T$, $\mathbf{A}^H$, $\|\mathbf{A}\|_F$, $\|\mathbf{A}\|_p$, $\text{Tr}(\mathbf{A})$, respectively. For a vector $\mathbf{x}$, denote its ℓ_2-norm by $\|\mathbf{x}\|_2$. And an identity matrix is denoted by $\mathbf{I}$. For a set F, denote its cardinality by $\mu(F)$.

B. HSI Data

HSI is obtained by the hyperspectral sensor which simultaneously images the target region with tens to hundreds of continuous and subdivided spectral bands in the ultraviolet, visible, near-infrared, and mid-infrared ranges of the electromagnetic spectrum. Therefore, HSIs in different bands represent the imaging results of the target in different electromagnetic frequency waves. Unfortunately, HSI is often contaminated by several different types of noise in the process of imaging, e.g., Gaussian noise, impulse noise, dead pixels, and stripes [11]. Analogy to [11], we model impulse noise, dead pixels, and stripes as sparse noise. As shown in Fig. 1, the observed HSI data with mixed-noise can be modeled as

$$\mathbf{O} = \mathbf{X} + \mathbf{S} + \mathbf{N} \tag{1}$$

where $\mathbf{O} = [\mathbf{O}_1, \mathbf{O}_2, \ldots, \mathbf{O}_p] \in \mathbb{R}^{MN\times p}$ represents the Casorati matrix of the degraded HSI $\mathcal{O} \in \mathbb{R}^{M\times N\times p}$, and $\mathbf{X}$, $\mathbf{S}$, and $\mathbf{N}$ denote the Casorati matrices of the clean image $\mathcal{X}$, the sparse noise image $\mathcal{S}$, and the Gaussian noise image $\mathcal{N}$, respectively.

C. HSI Restoration Problem Formulation

There have been various HSI restoration methods proposed for removing different kinds of noise in HSI, one representative approach is the LR matrix recovery (LRMR) method [11]. LRMR, a convex surrogate of robust principal component analysis (RPCA) model, is presented to reveal the LR structure of HSI by exploiting the strong correlation of the adjacent bands as well as the spatial adjacent pixels. The emergence of LRMR has greatly promoted the application of LR-based theory, which can be viewed as a baseline method for HSI restoration. Based on the observation model (1), the optimization problem of LRMR is formulated as

$$\begin{aligned} &\min_{\mathbf{X},\mathbf{S}\in\mathbb{R}^{MN\times p}} \|\mathbf{X}\|_* + \lambda\|\mathbf{S}\|_1 \\ &\text{s.t. } \|\mathbf{O}-\mathbf{X}-\mathbf{S}\|_F^2 \le \varepsilon, \quad \text{rank}(X) \le r \end{aligned} \tag{2}$$

where λ is the regularization parameter used to trade off nuclear norm and ℓ_1-norm, Frobenius norm is utilized to constrain the Gaussian noise, rank() term is used to pilot the number of nonzero singular values in singular value decomposition (SVD) when we resolve the nuclear norm, ε is a small tolerance, and r is a positive integer. Model (2) exploits the low-dimensional structure in high-dimensional data and has the ability to remove the Gaussian noise with small amplitude. However, this model is sensitive to outliers and is difficult to achieve good performance when the HSI is corrupted by mixed-noise [23].

III. HSI Representation by Graph Signal Processing

A. Fundamentals of GSP

An undirected graph is defined as $\mathcal{G} := (V, E, \mathbf{A})$ where $V = \{1, 2, \ldots, N\}$ encompasses N nodes on graph, E is the set of edges, and $\mathbf{A}$ is the weighted adjacency matrix of the graph and its entries $a_{i,j}$ denote the weight of the edge linking nodes i and j for $i, j \in V$. The degree matrix $\mathbf{D}$ of graph $\mathcal{G}$ is a diagonal matrix whose diagonal entries $d_{i,i}$ are equal to the sum of the ith row elements of the weighted adjacency matrix $\mathbf{A}$. The combinatorial Laplacian matrix $\mathbf{L}_\mathcal{G}$ is defined as $\mathbf{L}_\mathcal{G} = \mathbf{D}-\mathbf{A}$, and the normalized Laplacian matrix is defined as $\mathbf{L}_{\text{nor}} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}$. The eigendecomposition of $\mathbf{L}_\mathcal{G}$ is given by $\mathbf{L}_\mathcal{G} = \mathbf{U}_\mathcal{G}\Lambda\mathbf{U}_\mathcal{G}^T$, where $\mathbf{U}_\mathcal{G} = [\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_N]$ is an orthogonal matrix consisted of all eigenvectors, and $\Lambda = \text{diag}(\lambda_1, \lambda_2, \ldots, \lambda_N)$ is a diagonal matrix composed of all eigenvalues [30], [46], [47].

For a graph $\mathcal{G}$, the geodesic distance $\rho(i, j)$ is defined as the number of edges in the shortest path between nodes i and j, and the d-neighborhood set of node $i \in V$ is defined as $B_\mathcal{G}(i, d) := \{j \in V : \rho(i, j) \le d\}$ for $i, j \in V$ [44].

In GSP, the signal resided on the graph $\mathcal{G}$ is represented as a vector $\mathbf{x} = [x_1, x_2, \ldots, x_N]^T$, where the element x_i is indexed by the node $i \in V$. The graph Fourier transform of the signal is defined by $\hat{\mathbf{x}} = \mathbf{U}_\mathcal{G}^T\mathbf{x} = [\hat{x}_{\lambda_1}, \hat{x}_{\lambda_2}, \ldots, \hat{x}_{\lambda_N}]^T$ and the inverse transform is represented by $\mathbf{x} = \mathbf{U}_\mathcal{G}\hat{\mathbf{x}}$ [30], [47].

B. GSP Representation for HSI

GSP is a promising framework to represent, analyze and process datasets, where the physical proximity and correlation between the data elements can be captured by graph [31], [48]. As we know, HSI is a geometric structure data, thus the similarity between HSI bands and the intraband PWS characteristic can be efficiently modeled by the graph structure.

Similar to our previous work in [49], we model each band of HSI with a size of $M \times N \times p$ as a gray-level image, then the intensities of the pixels in gray-level image are modeled as the graph signal. In specific, pixels are represented by nodes, and any two adjacent pixels are connected by edges. Then the

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weights of the edges are described by a weighted adjacency matrix $\mathbf{A}$. Without loss of generality, the elements of $\mathbf{A}$ are initially computed using a Gaussian kernel, similar to that in [31]

$$a_{i,j} = \exp\left(-\frac{\|\mathbf{I}_i - \mathbf{I}_j\|_2^2}{\sigma_l^2}\right)\exp\left(-\frac{(x_i - x_j)^2}{\sigma_x^2}\right) \tag{3}$$

where $\mathbf{I}_i$ and x_i are the location and pixel value of pixel i in one band, respectively, ${\sigma_l}^2$ and ${\sigma_x}^2$ are two parameters. It is observed that the edge weight is not only related to the geometric distance between nodes (pixels), but also correlates with the pixel values, which is an adaptive manner for describing the similarity. From the computation perspective, the graph is constructed to be sparse, thus the d-neighborhood set is incorporated to the Gaussian kernel to define the matrix $\mathbf{A}$. In specific, when $\rho(i, j) > d$, weight $a_{i,j} = 0$. If we set $d = 1$, then each pixel only connects with its north, south, east, west (NWSE) neighbors and the corresponding edge weights are evaluated by (3).

For simplicity, we use the same graph, called as the detailed graph, for all bands of HSI, which is represented by $\mathcal{G}_d := (V_d, E_d, \mathbf{A}_d)$, where $V_d = \{1, 2, \ldots, MN\}$ denotes the nodes set consisting of the MN pixels of one band image, E_d represents the edges set, and $\mathbf{A}_d$ is evaluated by (3). Now the pixel values of each band of HSI can be modeled as a signal on the graph $\mathcal{G}_d$ and signals of all bands can be collected into a matrix $\mathbf{X} = [X(i, b)] \in \mathbb{R}^{MN\times p}$, where $X(i, b)$ denotes the graph signal of ith node in band b.

Besides, graph signals tend to be smooth on graph in real-world datasets, which is also suitable for the graph signal of HSI [42]. The smoothness of graph signals is a qualitative property, which can qualitatively indicate the frequency variability of graph signals on the graph [50]. The most commonly used smoothness indicator is GLR [42], [51]. Given a graph signal $\mathbf{x} \in \mathbb{R}^N$ resided on the $\mathcal{G}$ with Laplacian matrix $\mathbf{L}_\mathcal{G}$, the GLR can be expressed as [51]

$$\|\mathbf{x}\|_{\text{GLR}} = \mathbf{x}^T\mathbf{L}_\mathcal{G}\mathbf{x} = \sum_{i\sim j} a_{i,j}\cdot(x_i - x_j)^2 = \sum_{k=1}^{MN}\lambda_k\hat{x}_{\lambda_k}^2. \tag{4}$$

And the GLR of signals of all bands of HSI is defined as $\|\mathbf{X}\|_{\text{GLR}} = \text{Tr}(\mathbf{X}^T\mathbf{L}_\mathcal{G}\mathbf{X}) = \|\mathbf{x}_1\|_{\text{GLR}} + \cdots + \|\mathbf{x}_b\|_{\text{GLR}} + \cdots + \|\mathbf{x}_p\|_{\text{GLR}}$, which can be considered as a combination vector form of GLR. GLR is a quadratic form of matrix with lots of remarkable properties, in which the solution to the quadratic problem is not hard to obtain by standard linear algebra approaches [52].

C. Superpixel Segmentation and Skeleton Graph Construction

As aforementioned, we employ a two-layer graph structure to characterize the HSI. The lower layer, called the detailed graph, has been introduced in Section III-B. Now, we define the upper layer, called the skeleton graph constructed by superpixels of HSI. As we know, a superpixel is a small area composed of a series of pixels with adjacent positions and similar characteristics such as color, luminance, and texture [53], which retains the effective boundary information. By superpixel segmentation, the image features can be roughly expressed by a small number of superpixels, instead of a large number of pixels, which reduces the complexity of image processing in image preprocessing [54], image compression [55], image foreground extraction [56], and HSI unmixing [57].



(a) (b)

Fig. 2. (a) Original HSI. (b) Original HSI segmented by SLIC algorithm.

There are various algorithms for generating superpixels, one of which is the simple linear iterative clustering (SLIC) method, which has high computing and memory efficiency and exhibits the state-of-the-art boundary adherence [53]. A detailed description of SLIC can be found in [53] and [54], and an associated toolbox called VLFeat is shared in [58]. In the proposed algorithm, SLIC is applied to segment HSI for one band into a series of superpixels, and an example is shown in Fig. 2. It can be seen that SLIC achieves excellent performance in boundary information preservation.

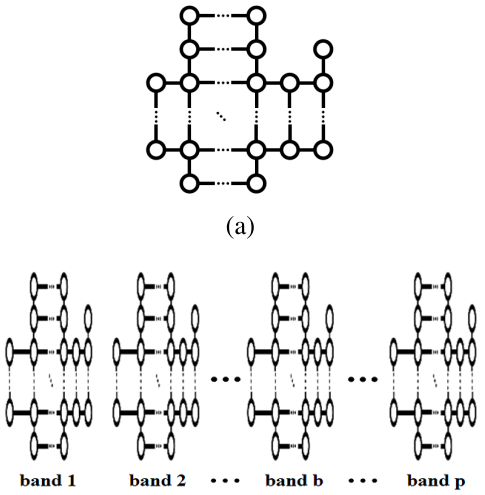
Based on the superpixels, we can define the skeleton graph for the HSI. Similar to that of detailed graph, we also use the same skeleton graph for every band of HSI. Suppose that there are M_s superpixels using superpixel segmentation for one band of HSI. Then, a modified KNN algorithm is employed to construct the skeleton graph

$$\hat{\mathcal{G}}_s := \left(\hat{V}_s, \hat{E}_s, \hat{\mathbf{A}}_s\right) \tag{5}$$

where $\hat{V}_s = \{1, 2, \ldots, M_s\}$ encompasses all nodes each of which corresponds to one superpixel and $\hat{E}_s$ denotes the edge set linking the nodes. For the skeleton graph, the mth entry of the signal $\mathbf{x}_s \in \mathbb{R}^{M_s}$ is defined as the average value of the values of the pixels belonging to the mth superpixel for $m = 1, 2, \ldots, M_s$. And the coordinates of mth node $v \in \hat{V}_s$ is defined as coordinate of the center pixel whose intensity is closest to the corresponding graph signal. With the signal and coordinates, we define the adjacency matrix $\hat{\mathbf{A}}_{\text{temp}}$ for the skeleton graph, whose entries are calculated by (3). To make the adjacency matrix sparse and symmetric, we first set the k largest elements in each row of $\hat{\mathbf{A}}_{\text{temp}}$ to be 1, and the other elements of each row to be 0. The adjacency matrix is then symmetrized and modified into unweighted one, the resultant adjacency matrix of the skeleton graph is finally denoted by $\hat{\mathbf{A}}_s$.

We close this section by defining the local graphs, used to formulate the HSI restoration problem. Based on the superpixel segmentation, the HSI is segmented into M_s superpixels each of which encompasses a set of original pixels (nodes). To capture the correlations between these pixels (nodes) within each superpixel, we define a series of graphs, called local graphs, each of which corresponds to one superpixel. For

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(a)

(b)

Fig. 3. (a) Grid graph of one superpixel. (b) Graph representation of the superpixel of HSI.

convenience, define the local graph associated with the mth superpixel with K_m pixels as

$$\tilde{\mathcal{G}}_m = \left(\tilde{V}_m, \tilde{E}_m, \tilde{\mathbf{A}}_m\right) \tag{6}$$

for $m = 1, 2, \ldots, M_s$, where $\tilde{V}_m$ encompasses K_m nodes on graph, $\tilde{E}_m$ is the set of edges and $\tilde{\mathbf{A}}_m$ represents the similarities between pixels. Then the graph signal resided on the local graph $\tilde{\mathcal{G}}_m$ is denoted as $\mathbf{x}_m = [x_{m,1}, x_{m,2}, \ldots, x_{m,K_m}]^T \in \mathbb{R}^{K_m}$. For illustration, one example of the local graph is shown in Fig. 3(a). For consistency, we use the same set of graphs for all bands of the same superpixel [shown in Fig. 3(b)], while the graph signals corresponding to the pixel values differ in different bands. Then, the graph signal representation of the bth band of superpixel m can be defined as a vector $\mathbf{X}_{m,b} = [X_m(1, b), X_m(2, b), \ldots, X_m(K_m, b)]^T \in \mathbb{R}^{K_m}, b = 1, \ldots, p$, where $X_m(i, b)$ denotes the graph signal value of the ith node of local graph $\tilde{\mathcal{G}}_m$ in the bth band. Thus, the sequential signals on local graph $\tilde{\mathcal{G}}_m, m = 1, 2, \ldots, M_s$ can be obtained by collecting the graph signals of all bands into a matrix, denoted by $\mathbf{X}_m = [X_m(i, b)] = [\mathbf{X}_{m,1}, \ldots, \mathbf{X}_{m,b}, \ldots, \mathbf{X}_{m,p}] \in \mathbb{R}^{K_m \times p}$.

Remark 1: By the local graphs, the detailed graph $\mathcal{G}_d$ can also be defined as the combination of the disjoint local graphs, i.e., $\mathcal{G}_d = \cup_{m=1}^{M_s} \tilde{\mathcal{G}}_m$, and when the boundary of the two superpixels corresponding to the local graph is neighboring in original HSI, the boundary nodes will connect two local graphs.

IV. Proposed Method

In this section, the GLRSSDA model is proposed for HSI restoration. The diagram of proposed method is shown in Fig. 4. First, a quick denoising algorithm with low complexity is applied for one band of the noisy HSI. Second, the denoised HSI band is segmented into a series of superpixels. Then, each superpixel is modeled as a node, and the similarities between nodes are represented by the edges, thus the underlying structure of the intercorrelation between superpixels, which includes the nonlocal similarities, can be characterized by a skeleton graph $\hat{\mathcal{G}}_s$. Finally, based on the skeleton graph, all bands of the original noisy HSI are restored in a distributed framework. In distributed framework, the restoration is performed on the nodes of skeleton graph using the information on several local graphs, according to certain nonlocal interaction scheme. The process of the proposed algorithm is elaborated below.

Remark 2: Different from the existing HSI restoration method, which formulates the restoration problem into a global optimization problem, the proposed method restores the HSI in a distributed framework, which formulates the restoration problem into a series of optimization problems with small scales. Accordingly, the proposed method can relieve the heavy burden of the computing unit by solving the problems locally, which makes it suitable for HSI with large size.

A. Preprocessing

Denoising is the first step in the proposed method, because the HSI with heavy noise will have an adverse impact on superpixel segmentation. Thus, a fast and efficient image denoising method is chosen for one band of HSI. Note that this band can be selected either specifically or randomly, the band of the proposed method is selected as 50 for all the dataset (the selection will not influence heavily on the performance, since we will perform denoising on the image of the band before constructing the graphs). For a quick deployment, a median filter is applied to reduce the noise of HSI in the proposed method, while the remaining noise will have small impact on superpixel segmentation [59]. There are also other candidate denoising methods, such as the spectral average method [29] and the first PCA method [60].

B. Skeleton Graph Construction

After preprocessing, the denoised image is segmented into a series of superpixels by the SLIC algorithm [53], then the skeleton graph $\hat{\mathcal{G}}_s$ is constructed by a modified KNN algorithm, which is detailed in Section III-C. Note that the superpixel segmentation and skeleton graph $\hat{\mathcal{G}}_s$ construction are handled in the selected band of HSI. Due to the high correlations between HSI bands, it is reasonable to use the skeleton graph to represent the intercorrelations between superpixels for all bands.

C. Deployment of Distributed Algorithm

In this article, we devise a distributed algorithm for restoring the HSI region by region. Before describing the algorithm, we define the notion of subgraph each of which corresponds to a region. And the restoration problem is then formulated as a series of problems with small scales each of which resides on one subgraph. Based on the local graphs (6), we define a series of overlapping subgraphs each of which is centered at one node on $\hat{\mathcal{G}}_s$

$$\bar{\mathcal{G}}_f = \left(\bar{V}_f, \bar{E}_f, \bar{\mathbf{A}}_f\right), \quad f = 1, 2, \ldots, M_s \tag{7}$$

where $\bar{V}_f = \{\tilde{V}_m\}_{m \in B_{\hat{\mathcal{G}}_s}(f,d)}$ and $\bar{E}_f$ denotes the edge set, defined later. It is observed that the subgraph consists of mth local graphs with $m \in B_{\hat{\mathcal{G}}_s}(f, d)$. Now we define the

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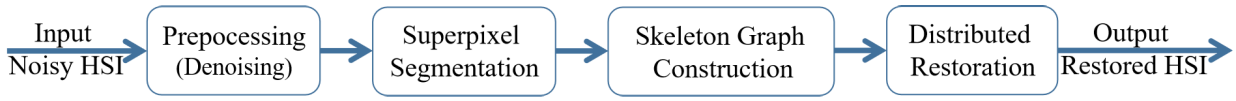


Fig. 4. Diagram of proposed method.

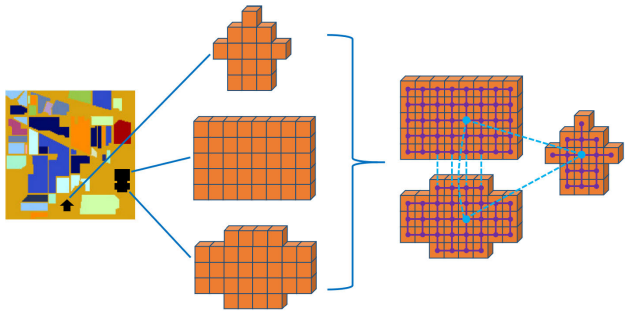


Fig. 5. Example of the construction of subgraph.

edges of $\bar{\mathcal{G}}_f$. If the local graph $\tilde{\mathcal{G}}_f$ and $\tilde{\mathcal{G}}_m$ are connected, then the edges linking the nodes from these local graphs inherit the ones in the detailed graph. Otherwise, there is only one edge between the disconnected local graphs, i.e., there is only one edge between the center pixels of two local graphs. An example of the construction of subgraph is shown in Fig. 5. In specific, suppose that there are three superpixels (three black areas in Fig. 5 with irregular shapes), each of which corresponds to a local graph. The edges of each local graph are represented by purple lines. Besides, suppose that the three local graphs belong to one subgraph, there are edges connections between the center pixels of every two superpixels. Moreover, it can be seen from Fig. 5 that the rectangular region and the dodecagonal region are adjacent, thus the adjacent pixels are connected by edges. These edges are represented by blue dotted lines in Fig. 5. Accordingly, the Laplacian matrix of $\bar{\mathcal{G}}_f$ can be obtained, denoted as $\mathbf{L}_{\bar{\mathcal{G}}_f}$. Note that we use the same subgraphs for all bands of the HSI region corresponding to the subgraph. The signal on the fth subgraph is formed by stacking the values of all pixels $v \in \bar{V}_f$, i.e., $\mathbf{X}_{f,b} = [X_f(1,b), X_f(2,b), \ldots, X_f(\sum K_m, b)]^T \in \mathbb{R}^{\sum K_m}$, where $m \in B_{\hat{\mathcal{G}}_s}(f,d)$, $b = 1, \ldots, p$, and $X_f(i,b)$ denotes the graph signal value of the ith node of the subgraph $\bar{\mathcal{G}}_f$ in the bth band. Then, the sequential signals on the subgraph $\bar{\mathcal{G}}_f$ can be obtained by collecting the graph signals of all bands into a matrix, i.e., $\mathbf{X}_f = [X_f(i,b)] = [\mathbf{X}_{f,1}, \ldots, \mathbf{X}_{f,b}, \ldots, \mathbf{X}_{f,p}] \in \mathbb{R}^{(\sum K_m)\times p}$, $m \in B_{\hat{\mathcal{G}}_s}(f,d)$. From the definition of (7), it can be seen that the larger d is, the more neighbors the node has, so it provides more similarity information for the node, and also causes greater computational cost. Thus, a trade-off between similarity information and computational costs needs to be considered. In the proposed algorithm, the node set of each subgraph is determined by $B(i,d)$ with $d = 1$.

Then a tailored distributed manner is applied in each subgraph, which is an iterative scheme. Suppose that there are M_s computing units, each of which corresponds to a subgraph $\bar{\mathcal{G}}_f$. And each unit establishes transmission connections with its neighbor computing unit based on the edges of the skeleton graph $\hat{\mathcal{G}}_s$. In each iteration, each computing unit resolves one restoration problem and exchanges the result with its neighbors. As a result, each pixels will achieve more than one restoration values, which is then averaged. The averaged pixels values will then be fed into the next iteration. Such distributed scheme is also called the divide-and-conquer strategy [61].

The computing model of each computing unit, corresponding to each subproblem, is detailed in Section IV-D. Note that the M_s computing units work simultaneously to solve the model, a parallel scheme. And one computing unit in physical can correspond to several computing units in the proposed model.

D. Optimization Algorithm for Each Subproblem

In this section, we will discuss the restoration of HSI region by region. For each region, we restore the pixel values of nodes residing on one subgraph, where the restoration problem is formulated into a constrained optimization problem. Therefore, we formulate the restoration problem of the entire HSI into a series of subproblems (note that we will no longer formulate the restoration problem into a global problem in this distributed approach). The fth problem corresponding to the subgraph $\bar{\mathcal{G}}_f$ is shown as

$$\min_{\mathbf{X}_f, \mathbf{S}_f \in \mathbb{R}^{(\sum K_m)\times p}} \|\mathbf{X}_f\|_* + \alpha \|\mathbf{X}_f\|_{\mathrm{GLR}} + \beta \|\mathbf{S}_f\|_1 \quad \text{s.t. } \|\mathbf{O}_f - \mathbf{X}_f - \mathbf{S}_f\|_F^2 \le \varepsilon, \quad \mathrm{rank}(\mathbf{X}_f) \le r \tag{8}$$

where $m \in B_{\hat{\mathcal{G}}_s}(f,d)$, K_m represents the number of the pixels in the mth superpixel, α and β are two parameters used to get trade-off between the three terms of the objective function. $\mathbf{O}_f$ and $\mathbf{S}_f$ represent the observed noisy signal and sparse noise corresponding to $\mathbf{X}_f$, respectively. The problem (8) is nonconvex due to the rank constraint. In this article, we employ the augmented Lagrangian multiplier (ALM) method [62], [63] to solve the problem.

First of all, the problem (8) is converted to the following equivalent problem by introducing auxiliary variable $\mathbf{H}_f$:

$$\min_{\mathbf{H}_f, \mathbf{X}_f, \mathbf{S}_f \in \mathbb{R}^{(\sum K_m)\times p}} \|\mathbf{H}_f\|_* + \alpha \|\mathbf{X}_f\|_{\mathrm{GLR}} + \beta \|\mathbf{S}_f\|_1 \quad \text{s.t. } \|\mathbf{O}_f - \mathbf{H}_f - \mathbf{S}_f\|_F^2 \le \varepsilon, \quad \mathrm{rank}(\mathbf{H}_f) \le r, \ \mathbf{H}_f = \mathbf{X}_f. \tag{9}$$

Then the augmented Lagrangian function of problem (9) is expressed as

$$\begin{aligned} &\ell(\mathbf{H}_f, \mathbf{X}_f, \mathbf{S}_f, \mathbf{C}_f, \mathbf{D}_f) \\ &= \|\mathbf{H}_f\|_* + \alpha \|\mathbf{X}_f\|_{\mathrm{GLR}} + \beta \|\mathbf{S}_f\|_1 \\ &\quad + \langle \mathbf{C}_f, \mathbf{O}_f - \mathbf{H}_f - \mathbf{S}_f \rangle + \langle \mathbf{D}_f, \mathbf{X}_f - \mathbf{H}_f \rangle \\ &\quad + \frac{\gamma}{2}\left(\|\mathbf{O}_f - \mathbf{H}_f - \mathbf{S}_f\|_F^2 + \|\mathbf{X}_f - \mathbf{H}_f\|_F^2\right) \\ &\quad \text{s.t. } \mathrm{rank}(\mathbf{H}_f) \le r \end{aligned} \tag{10}$$

where γ ($\gamma > 0$) is the penalty parameter, $\mathbf{C}_f$ and $\mathbf{D}_f$ are the Lagrange multipliers. We can alternatively optimize the variables in (10), in which one variable is optimized while others are fixed. Therefore, in the $(k+1)$th iteration, variables in the problem (10) can be updated specifically as follows:

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1) Update $\mathbf{H}_f$*:* Extracting all terms containing $\mathbf{H}_f$ from (10), we can deduce that

$$\begin{aligned}\mathbf{H}_f^{(k+1)} &= \underset{\text{rank}(\mathbf{H}_f)\le r}{\text{argmin}}\ \ell\left(\mathbf{H}_f, \mathbf{X}_f^{(k)}, \mathbf{S}_f^{(k)}, \mathbf{C}_f^{(k)}, \mathbf{D}_f^{(k)}\right)\\ &= \underset{\text{rank}(\mathbf{H}_f)\le r}{\text{argmin}}\ \|\mathbf{H}_f\|_* + \langle \mathbf{C}_f^{(k)}, \mathbf{O}_f - \mathbf{H}_f - \mathbf{S}_f^{(k)}\rangle\\ &\quad + \langle \mathbf{D}_f^{(k)}, \mathbf{X}_f^{(k)} - \mathbf{H}_f\rangle + \frac{\gamma}{2}\|\mathbf{O}_f - \mathbf{H}_f - \mathbf{S}_f^{(k)}\|_F^2\\ &\quad + \frac{\gamma}{2}\|\mathbf{X}_f^{(k)} - \mathbf{H}_f\|_F^2\\ &= \underset{\text{rank}(\mathbf{H}_f)\le r}{\text{argmin}}\ \|\mathbf{H}_f\|_* + \gamma\|\mathbf{H}_f - \frac{1}{2}\Big(\mathbf{O}_f + \mathbf{X}_f^{(k)}\\ &\qquad - \mathbf{S}_f^{(k)} + \frac{1}{\gamma}\Big(\mathbf{C}_f^{(k)} + \mathbf{D}_f^{(k)}\Big)\Big)\|_F^2. \end{aligned} \tag{11}$$

To solve the problem (11), a Lemma is introduced as [64].

Lemma 1: Consider the SVD of a matrix $\mathbf{Z} \in \mathbb{R}^{K\times p}$ of rank r

$$\mathbf{Z} = \mathbf{U}\mathbf{E}_r\mathbf{V}^H, \quad \mathbf{E}_r = \text{diag}(\sigma_i) \tag{12}$$

where $\mathbf{U} \in \mathbb{R}^{K\times r}$ and $\mathbf{V} \in \mathbb{R}^{p\times r}$ are orthogonal matrices, and the singular values $\sigma_{i\,(1\le i\le r)} \ge 0$. For a threshold $\phi \ge 0$, the singular value thresholding (a.k.a. singular value shrinkage) operator $\mathcal{D}_\phi(\mathbf{Z})$ is defined as

$$\begin{aligned}\mathcal{D}_\phi(\mathbf{Z}) &:= \mathbf{U}\mathcal{D}_\phi(\mathbf{E}_r)\mathbf{V}^H\\ \mathcal{D}_\phi(\mathbf{E}_r) &= \text{diag}(\max(\eta, 0))\end{aligned} \tag{13}$$

where $\eta = \sigma_i - \phi$. The results of the singular value thresholding operator (13) are the solution of the problem

$$\mathcal{D}_\phi(\mathbf{Z}) = \underset{\text{rank}(\mathbf{H})\le r}{\text{argmin}}\ \|\mathbf{H}\|_* + \frac{1}{2\phi}\|\mathbf{H} - \mathbf{Z}\|_F^2. \tag{14}$$

Using Lemma 1, the result of the optimization problem (11) is given by

$$\mathbf{H}_f^{(k+1)} = \mathcal{D}_{\frac{1}{2\gamma}}\left(\frac{1}{2}\left(\mathbf{O}_f + \mathbf{X}_f^{(k)} - \mathbf{S}_f^{(k)} + \frac{1}{\gamma}\left(\mathbf{C}_f^{(k)} + \mathbf{D}_f^{(k)}\right)\right)\right). \tag{15}$$

2) Update $\mathbf{X}_f$*:* Extracting all terms containing $\mathbf{X}_f$ from (10), we can derive that

$$\begin{aligned}\mathbf{X}_f^{(k+1)} &= \underset{\mathbf{X}}{\text{argmin}}\ \ell\left(\mathbf{H}_f^{(k+1)}, \mathbf{X}_f, \mathbf{S}_f^{(k)}, \mathbf{C}_f^{(k)}, \mathbf{D}_f^{(k)}\right)\\ &= \underset{\mathbf{X}_f}{\text{argmin}}\ \alpha\|\mathbf{X}_f\|_{\text{GLR}} + \langle \mathbf{D}_f^{(k)}, \mathbf{X}_f - \mathbf{H}_f^{(k+1)}\rangle\\ &\quad + \frac{\gamma}{2}\|\mathbf{X}_f - \mathbf{H}_f^{(k+1)}\|_F^2\\ &= \underset{\mathbf{X}_f}{\text{argmin}}\ \alpha\|\mathbf{X}_f\|_{\text{GLR}} + \frac{\gamma}{2}\|\mathbf{X}_f - \mathbf{H}_f^{(k+1)}\\ &\quad + \frac{\mathbf{D}_f^{(k)}}{\gamma}\|_F^2. \end{aligned} \tag{16}$$

For convenience, let $\mathbf{B}_f = \mathbf{H}_f^{(k+1)} - \mathbf{D}_B^{(k)}/\gamma = [\mathbf{b}_1, \ldots, \mathbf{b}_p]$. According to the matrix form of GLR in Section III-B, the problem (16) is equivalent to

$$\mathbf{X}_f^{(k+1)} = \underset{\mathbf{X}_f=[\mathbf{x}_1,\ldots,\mathbf{x}_p]}{\text{argmin}} \sum_{j=1}^{p}\left(\alpha\|\mathbf{x}_j\|_{\text{GLR}} + \frac{\gamma}{2}\|\mathbf{x}_j - \mathbf{b}_j\|_2^2\right). \tag{17}$$

Then, the problem (17) can be decoupled to p subproblems, which can be solved band by band. The jth subproblem of (17) is

$$\begin{aligned}\mathbf{x}_j^{(k+1)} &= \underset{\mathbf{x}_j}{\text{argmin}}\ \alpha\|\mathbf{x}_j\|_{\text{GLR}} + \frac{\gamma}{2}\|\mathbf{x}_j - \mathbf{b}_j\|_2^2\\ &= \alpha\mathbf{x}_j^T\mathbf{L}_{\bar{\mathcal{G}}_f}\mathbf{x}_j + \frac{\gamma}{2}\|\mathbf{x}_j - \mathbf{b}_j\|_2^2 \end{aligned} \tag{18}$$

where $\mathbf{L}_{\bar{\mathcal{G}}_f}$ is the combinatorial Laplacian matrix of $\bar{\mathcal{G}}_f$. The problem of (18) is a least squares problem and it admits a closed form solution

$$\mathbf{x}_j^{(k+1)} = \left(2\alpha\mathbf{L}_{\bar{\mathcal{G}}_f} + \gamma\mathbf{I}\right)^{-1}\gamma\mathbf{b}_j. \tag{19}$$

Thus, the solution of (17) can be obtained by combining the solution of each band of (19)

$$\begin{aligned}&\left[\mathbf{x}_1^{(k+1)}, \ldots, \mathbf{x}_f^{(k+1)}, \ldots, \mathbf{x}_p^{(k+1)}\right]\\ &= \left[\left(2\alpha\mathbf{L}_{\bar{\mathcal{G}}_f} + \gamma\mathbf{I}\right)^{-1}\gamma\mathbf{b}_1\right.\\ &\qquad \left., \ldots, \left(2\alpha\mathbf{L}_{\bar{\mathcal{G}}_f} + \gamma\mathbf{I}\right)^{-1}\gamma\mathbf{b}_f, \ldots, \left(2\alpha\mathbf{L}_{\bar{\mathcal{G}}_f} + \gamma\mathbf{I}\right)^{-1}\gamma\mathbf{b}_p\right]\\ &= \left(2\alpha\mathbf{L}_{\bar{\mathcal{G}}_f} + \gamma\mathbf{I}\right)^{-1}\left[\mathbf{b}_1, \ldots, \mathbf{b}_f, \ldots, \mathbf{b}_p\right]. \end{aligned} \tag{20}$$

Apparently, problem (20) can be rewritten as a matrix form

$$\mathbf{X}_f^{(k+1)} = \left(2\alpha\mathbf{L}_{\bar{\mathcal{G}}_f} + \gamma\mathbf{I}\right)^{-1}\mathbf{B}_f. \tag{21}$$

Through deducing, the solution of the problem (16) can equivalently be expressed as

$$\begin{aligned}\mathbf{X}_f^{(k+1)} &= \underset{\mathbf{X}_f}{\text{argmin}}\ \alpha\|\mathbf{X}_f\|_{\text{GLR}} + \frac{\gamma}{2}\|\mathbf{X}_f - \mathbf{H}_f^{(k+1)}\\ &\quad + \frac{\mathbf{D}_f^{(k)}}{\gamma}\|_F^2\\ &= \underset{\mathbf{X}_f}{\text{argmin}}\ \alpha\text{Tr}\left(\mathbf{X}_f^T\mathbf{L}_{\bar{\mathcal{G}}_f}\mathbf{X}_f\right) + \frac{\gamma}{2}\|\mathbf{X}_f - \mathbf{H}_f^{(k+1)} + \frac{\mathbf{D}_f^{(k)}}{\gamma}\|_F^2\\ &= \left(2\alpha\mathbf{L}_{\bar{\mathcal{G}}_f} + \gamma\mathbf{I}\right)^{-1}\left(\mathbf{H}_f^{(k+1)} - \frac{\mathbf{D}_f^{(k)}}{\gamma}\right). \end{aligned} \tag{22}$$

3) Update $\mathbf{S}_f$*:* Extracting all terms containing $\mathbf{S}_f$ from (10), we can deduce that

$$\begin{aligned}\mathbf{S}_f^{(k+1)} &= \underset{\mathbf{S}_f}{\text{argmin}}\ \ell\left(\mathbf{H}_f^{(k+1)}, \mathbf{X}_f^{(k+1)}, \mathbf{S}_f, \mathbf{C}_f^{(k)}, \mathbf{D}_f^{(k)}\right)\\ &= \underset{\mathbf{S}_f}{\text{argmin}}\ \beta\|\mathbf{S}_f\|_1 + \langle \mathbf{C}_f^{(k)}, \mathbf{O}_f - \mathbf{H}_f^{(k+1)} - \mathbf{S}_f\rangle\\ &\quad + \frac{\gamma}{2}\|\mathbf{O}_f - \mathbf{H}_f^{(k+1)} - \mathbf{S}_f\|_F^2\\ &= \underset{\mathbf{S}_f}{\text{argmin}}\ \beta\|\mathbf{S}_f\|_1 + \frac{\gamma}{2}\|\mathbf{S}_f\\ &\quad - \left(\mathbf{O}_f - \mathbf{H}_f^{(k+1)} + \frac{\mathbf{C}_f^{(k)}}{\gamma}\right)\|_F^2. \end{aligned} \tag{23}$$

By introducing the following soft-thresholding operator:

$$\mathcal{R}_\triangle(\varphi) = \begin{cases}\varphi - \triangle, & \text{if } \varphi > \triangle\\ \varphi + \triangle, & \text{if } \varphi < \triangle\\ 0, & \text{otherwise}\end{cases} \tag{24}$$

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Algorithm 1 GLRSSDA Algorithm

Inputs: Observed noisy signal $\mathcal{O}$, desired rank r, the regularized parameters α, and β.

1. Denoise one band of $\mathcal{O}$.

2. Segment the result of step 1 into M_s superpixels and then construct skeleton graph via the steps in Section III-C.

3. Based on the skeleton graph, formulate the restoration of entire HSI into a series of optimization problems.

4. Solve each problem resided on the corresponding subgraph $\bar{\mathcal{G}}_f$ in a distributed manner, $f = 1, \ldots, M_s$**:**

Initialization: $\mathbf{H}_f = \mathbf{X}_f = \mathbf{S}_f = \mathbf{C}_f = \mathbf{D}_f = \mathbf{0}$, graph Laplacian matrix $\mathbf{L}_{\bar{\mathcal{G}}_f}$, initial penalty parameter $\gamma = 10^{-2}$ and its maximum $\gamma_{max} = 10^6$, growth coefficient of penalty coefficient $\rho = 1.5$, iteration $k = 0$, and tolerance $\varepsilon_1 = \varepsilon_2 = 10^{-6}$.

1) Update $\mathbf{H}_f^{k+1}$, $\mathbf{X}_f^{k+1}$, $\mathbf{S}_f^{k+1}$, $\mathbf{C}_f^{k+1}$, $\mathbf{D}_f^{k+1}$ via equations (15), (22), (25), and (26), respectively.

2) Send the intensities of $\mathbf{X}_f^{k+1}$, $\mathbf{H}_f^{k+1}$, $\mathbf{S}_f^{k+1}$, $\mathbf{C}_f^{k+1}$, $\mathbf{D}_f^{k+1}$ to its corresponding neighbors that are the nodes of $\tilde{V}_m$, $m \in B_{\hat{\mathcal{G}}_s}(f, d)$, $m \neq f$.

3) Fuse and average the components of $\mathbf{H}_f$, $\mathbf{X}_f$, $\mathbf{S}_f$, $\mathbf{C}_f$, $\mathbf{D}_f$ that are resided on the node $\tilde{V}_f$ of the local graph $\tilde{\mathcal{G}}_f$, denoted as $\grave{\mathbf{Z}}_f = [\grave{\mathbf{H}}_f, \grave{\mathbf{X}}_f, \grave{\mathbf{S}}_f, \grave{\mathbf{C}}_f, \grave{\mathbf{D}}_f]^T$, to update the corresponding part of $\mathbf{H}_f^{k+1}$, $\mathbf{X}_f^{k+1}$, $\mathbf{S}_f^{k+1}$, $\mathbf{C}_f^{k+1}$, $\mathbf{D}_f^{k+1}$ by:

$$\grave{\mathbf{Z}}_f^{k+1} = \frac{1}{\mu\left(B_{\hat{\mathcal{G}}_s}(f, d)\right)} \sum_{m \in B_{\hat{\mathcal{G}}_s}(f,d)} \grave{\mathbf{Z}}_{f,m}^{k+1}$$

where $\grave{\mathbf{Z}}_{f,m}$ denotes the value of $\grave{\mathbf{Z}}_f$ which is sent (computed) by the neighbor subgraph $\bar{\mathcal{G}}_m$.

4) Send the components of $\mathbf{X}_f^{k+1}$, $\mathbf{H}_f^{k+1}$, $\mathbf{S}_f^{k+1}$, $\mathbf{C}_f^{k+1}$, $\mathbf{D}_f^{k+1}$ resided on the node $\tilde{V}_f$, i.e., $\grave{\mathbf{H}}_f^{k+1}$, $\grave{\mathbf{X}}_f^{k+1}$, $\grave{\mathbf{S}}_f^{k+1}$, $\grave{\mathbf{C}}_f^{k+1}$, $\grave{\mathbf{D}}_f^{k+1}$ to its neighbors $j \in \tilde{V}_f$.

5) Check the iteration stopping condition:

If $(\|\mathbf{O}_f - \mathbf{H}_f^{(k+1)} - \mathbf{S}_f^{(k+1)}\|_F / \|\mathbf{O}_f\|_F \leq \varepsilon_1$ && $\|\mathbf{H}_f^{(k+1)} - \mathbf{X}_f^{(k+1)}\|_\infty \leq \varepsilon_2)$

Break, and do not compute in next iteration.

Otherwise

Update parameter $\gamma := min(\rho\gamma, \gamma_{max})$, $k = k + 1$ and go to step 1).

5. Merge all the center of $\mathbf{X}_f$, $f = 1, \ldots, M_s$, then obtain the $\mathbf{X}$, which is nonoverlapping.

Output: Restored HSI $\mathbf{X}$.

where $\varphi \in \mathbb{R}$ and $\triangle > 0$, the optimization solution of (23) can be expressed as

$$\mathbf{S}_f^{(k+1)} = \mathcal{R}_{\frac{\beta}{\gamma}}\left(\mathbf{O}_f - \mathbf{H}_f^{(k+1)} + \frac{\mathbf{C}_f^{(k)}}{\gamma}\right). \tag{25}$$

4) Update Multipliers: Given $\mathbf{H}_f, \mathbf{X}_f, \mathbf{S}_f$, the multipliers are updated by the following equations:

$$\mathbf{C}_f^{(k+1)} = \mathbf{C}_f^{(k)} + \gamma\left(\mathbf{O}_f - \mathbf{H}_f^{(k+1)} - \mathbf{S}_f^{(k+1)}\right)$$

$$\mathbf{D}_f^{(k+1)} = \mathbf{D}_f^{(k)} + \gamma\left(\mathbf{X}_f^{(k+1)} - \mathbf{H}_f^{(k+1)}\right). \tag{26}$$

The proposed algorithm is summarized in Algorithm 1.

V. Experimental Results and Discussion

Both synthesis and real HSI data experiments are undertaken to demonstrate the effectiveness of the GLRSDA method for HSI restoration. To thoroughly evaluate the performance of the proposed algorithm, nine different noise removal models are chosen for comparison, i.e., LRMR [11], total-variation-regularized low-rank matrix factorization (LRTV) [21], total variation regularized low-rank tensor decomposition (LRTDTV) [22], NGmeet [25], GLF [18], SSSB [29], TSLRLN [15], adaptive weight graph total variation regularization andlow-rank matrix recovery (LRAWGTV) [49], and FGSLR [20].

In all of the following experiments, the parameters in these compared methods are manually adjusted according to their default strategies, and all of the experiments are run on MATLAB(R2018b) and a Windows 10 desktop with an Intel Core I7-7700 3.6-GHz processor and 64-GB RAM. Besides, to facilitate the numerical calculation and visualization, all bands of the HSI are normalized into [0, 1] and they will be stretched to the original level after restoration. In the simulation, the mean signal-to-noise ratio (MPSNR), mean structural similarity index measure (MSSIM), and Erreur Relative Globale Adimensionnelle de Synthese (ERGAS) are adopted to measure the restoration quality [22]. The larger MPSNR, MSSIM are, and the smaller ERGAS are, the better HSI restoration result is. The three quantitative indicators are defined as follows [11]:

$$\begin{aligned} \text{PSNR}_i &= 10 * \log_{10} \frac{MN}{\sum_{x=1}^{M}\sum_{y=1}^{N}\left(\hat{u}_i(x, y) - u_i(x, y)\right)^2} \\ \text{MPSNR} &= \frac{1}{p}\sum_{i=1}^{p} \text{PSNR}_i \end{aligned} \tag{27}$$

where u_i and $\hat{u}_i$ represent the reference image and the restored image in the ith band, respectively,

$$\begin{aligned} \text{SSIM}_i &= \frac{\left(2\mu_{u_i}\mu_{\hat{u}_i} + H_1\right)\left(2\sigma_{u_i\hat{u}_i} + H_2\right)}{\left(\mu_{u_i}^2 + \mu_{\hat{u}_i}^2 + H_1\right)\left(\sigma_{u_i}^2 + \sigma_{\hat{u}_i}^2 + H_2\right)} \\ \text{MSSIM} &= \frac{1}{p}\sum_{i=1}^{p} \text{SSIM}_i \end{aligned} \tag{28}$$

where μ_{u_i} and $\mu_{\hat{u}_i}$ denote the average values of image u_i and $\hat{u}_i$, σ_{u_i} and $\sigma_{\hat{u}_i}$ stand for the variances, and $\sigma_{u_i\hat{u}_i}$ is the covariance between u_i and $\hat{u}_i$. H_1, H_2 are two constants to avoid the instability when the denominator is close to 0. The third measurement is specified by

$$\text{ERGAS} = \sqrt{\frac{1}{p}\sum_{i=1}^{p}\frac{\text{mse}\left(u_i, \hat{u}_i\right)}{\text{Mean}_2(u_i)}} \tag{29}$$

where mse() is the MSE performance function and $\text{Mean}_2()$ denotes the average or mean of the matrix elements.

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A. Simulated Data Experiments

There are three synthetic datasets chosen for comparison. The first is Pavia Center [65] with the size of $1096 \times 715 \times 102$. Similar to that in [27], we remove some heavily noise-contaminated bands; a subimage of size $1096 \times 715 \times 80$ is used in the experiment. The second is a complete image of Washington DC Mall (WDC) [66] with the size of $1280 \times 307 \times 191$. And the last large dataset is a subregion of HYDICE image of WDC dataset with the size of $256 \times 256 \times 191$ [66].

To simulate noisy data, we added four different types of noise to the original synthetic HSI data. Both visual quality and quantitative measurements are adopted to evaluate the performance of all the restoration methods we select. The detailed cases are as follows.

1) *Case 1:* In this case, the noise intensity is equal for different bands. The same distribution of zero-mean Gaussian noise and the same percentage of impulse noise are added to each band. The variance of the Gaussian noise is $G = 0.025$, and the percentage of the impulse noise is $P = 0.05$.
2) *Case 2:* In this case, the noise intensity is equal for different bands. The same distribution of zero-mean Gaussian noise and the same percentage of impulse noise are added to each band. The variance of the Gaussian noise is $G = 0.1$, and the percentage of the impulse noise is $P = 0.2$.
3) *Case 3:* In this case, the noise intensity is different for different bands. That is, Gaussian noise of zero-mean and different variances is added to each band with the variance value randomly selected in the range $[0, 0.2]$, and different percentages of impulse noise are also randomly added from 0 to 0.2.
4) *Case 4:* In this case, Gaussian noise and impulse noise are added just as in case 3. In addition, similar to that in [22], the deadlines are added with the number of stripes randomly selected from 3 to 10, and the width of the deadlines is randomly generated from 1 to 3. Note that the deadlines are added from band 61 to band 100 for the WDC dataset and from band 39 to band 78 for the Pavia Center dataset, respectively.

Now, both visual quality and quantitative comparison are demonstrated as follows.

Tables I–III list the three quantitative indices comparison of different HSI restoration methods on different datasets. The best results among all methods are highlighted in bold in each case. From Tables I–III, it can be observed that the proposed method achieves the best restoration performance in most cases. Fig. 6 shows the denoising results of the methods in Pavia Center dataset under case 4. In the top of the enlarged box, it can be observed that all the compared methods loss the detailed information on the river, while the proposed GLRSSDA can preserve the detail texture owing to that it can effectively characterizes the underlying spatial correlation by the detailed graph and effectively exploits the information of neighbor superpixels by the distributed manner. Fig. 7 shows the denoising results of the methods in complete WDC dataset under case 4. All the methods can remove the noise, but some of them suffer from over-smoothing and losing detailed texture. Fig. 8 shows the denoising results of the methods in a subregion of WDC dataset under case 4. It verifies that the proposed method also exhibits superior performance in the dataset with small scale. It is clear that the LRTV and FGSLR lost effectiveness in the band. The NGmeet and SSSB lost effectiveness in some patches, due to losing information of neighbors. The LRAWGTV and GLRSSDA achieve the best denoising performance, and can preserve detailed information, such as the road detailed texture in the enlarged box, owing to the underlying structure characterization by GSP. It is worth mentioning that owing to the iterative distributed algorithm applied for solving the optimization problem, the proposed GLRSSDA can achieve better performance than LRAWGTV while using less neighbor information and costing less computing time. PSNR and SSIM values of each band of the three datasets in case 4 are shown in Fig. 9, it can be seen that the proposed GLRSSDA can steadily remove the noise in each band, while the FGSLR, TSLRLN, SSSB, LRTV, LRTDTV losing effectiveness in some bands. Besides, the noise intensity of different bands is generally different even in the same HSI, resulting in denoising performance fluctuations. In the fluctuation bands, our algorithm also shows excellent restoration performance, which can be observed from Fig. 9.

B. Real-World Data Experiments

In this section, we present the experimental results of two real-world datasets, i.e., HYDICE Urban dataset [67] and EO-1 Hyperion dataset [68].

1) HYDICE Urban Dataset [67]: The HYDICE Urban dataset is a common real dataset for demonstrating the performance of the restoration methods, whose size is $307 \times 307 \times 220$. As displayed in Fig. 10(b), the Urban dataset is corrupted by varieties of noise types, such as Gaussian, stripes, and deadlines.

Fig. 10 shows the denoising results for band 104 of Urban dataset via the different methods. It is clear that the restored HSIs by the methods are still corrupted by stripe noise, except those by the TSLRLN, LRAWGTV, and GLRSSDA methods. From Fig. 10(i) and the enlarged box, the TSLRLN can be regarded as enhancing the image, resulting in degradation of the original information. From the enlarged box in Fig. 10(j), the LRAWGTV suffers from a slight over-smoothing, losing detailed information. The vertical and horizontal mean profiles are also shown in Figs. 11 and 12, respectively, it also verifies that the proposed GLRSSDA achieves superior denoising performance.

2) EO-1 Hyperion Dataset [68]: The EO-1 Hyperion dataset is a large-scale dataset with size of $3466 \times 256 \times 242$. It can be observed from Fig. 13(b) that the dataset is corrupted by various noise, such as Gaussian and stripe noise. Figs. 13, 14, and 15 present the denoising results for band 119 of the dataset via the different methods. It can be observed that in Fig. 13, especially the enlarged box, the denoised HSI by the LRMR, LRTV, LRTDTV, GLF, LRAWGTV, and FGSLR still contain evident mixed-noise,

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TABLE I
QUANTITATIVE EVALUATION ON THE COMPLETE PAVIA CENTER DATASET IN DIFFERENT CASES

Noise case	Level	Evaluation index	LRMR	LRTV	LRTDTV	NGmeet	GLF	SSSB	TSLRLN	LRAWGTV	FGSLR	GLRSSDA
case 1	G=0.025 P=0.05	MPSNR	36.54	36.13	34.85	31.66	31 .98	32.42	32.02	40.29	36.10	**40.75**
		MSSIM	0.9787	0.9910	0.9879	0.9405	0.9416	0.9500	0.9441	0.9875	0.9206	**0.9970**
		ERGAS	63.15	66.24	76.47	118.11	107.76	104.83	107.30	88.59	302.57	**39.09**
		Times(s)	398.66	2202.4	2080.9	822.64	6561.7	4411.0	88.390	2275.5	39182	1024.5
case 2	G=0.1 P=0.2	MPSNR	28.67	29.41	29.33	20.62	21.13	27.44	21.21	32.10	27.21	**33.27**
		MSSIM	0.9006	0.9122	0.9137	0.7455	0.7456	0.8209	0.7651	0.9381	0.8045	**0.9772**
		ERGAS	155.61	143.44	143.51	422.69	381.40	182.24	379.01	213.81	691.09	**92.86**
		Times(s)	396.36	2327.3	1997.6	851.47	6436.4	9704.7	88.887	2113.5	39571	983.52
case 3	Gaussian +impulse	MPSNR	28.85	29.67	29.61	24.04	25.04	27.80	25.09	**33.49**	27.43	33.46
		MSSIM	0.9008	0.9162	0.9218	0.8129	0.8398	0.8343	0.8431	0.9676	0.8182	**0.9754**
		ERGAS	166.48	141.89	139.08	322.80	266.78	175.43	270.54	97.06	608.42	**92.71**
		Times(s)	387.12	2196.3	2060.7	873.68	6263.6	8949.4	88.096	2154.9	26305	984.59
case 4	Gaussian +impulse+ deadlines	MPSNR	28.82	29.62	29.57	24.10	25.11	27.88	25.12	**33.41**	27.42	33.35
		MSSIM	0.9001	0.9156	0.9219	0.8125	0.8394	0.8357	0.8425	0.9676	0.8183	**0.9758**
		ERGAS	166.64	143.15	139.81	321.68	265.44	173.69	269.35	97.57	607.94	**93.77**
		Times(s)	385.23	2200.9	2174.9	878.34	6662.2	8570.4	87.479	2184.2	26048	998.84

TABLE II
QUANTITATIVE EVALUATION ON THE COMPLETE WDC DATASET IN DIFFERENT CASES

Noise case	Level	Evaluation index	LRMR	LRTV	LRTDTV	NGmeet	GLF	SSSB	TSLRLN	LRAWGTV	FGSLR	GLRSSDA
case 1	G=0.025 P=0.05	MPSNR	39.24	43.86	45.67	31.10	31.15	38.72	31.17	48.56	48.58	**48.78**
		MSSIM	0.8005	0.9335	0.9645	0.6240	0.6258	0.8500	0.6283	**0.9867**	0.9538	0.9844
		ERGAS	304.55	72.27	62.59	773.12	777.22	208.68	780.94	42.12	336.19	**40.82**
		Times(s)	345.96	3652.4	2371.9	413.33	3193.4	364.91	185.69	2635.3	9929.1	1462.6
case 2	G=0.1 P=0.2	MPSNR	28.27	37.59	38.05	19.44	19.40	32.96	19.45	39.85	34.94	**39.92**
		MSSIM	0.4958	0.8047	0.8325	0.3863	0.3709	0.6519	0.3897	**0.8994**	0.6737	0.8849
		ERGAS	1183.4	157.32	200.71	3043.0	3021.9	474.12	3032.9	**139.49**	1758.3	140.35
		Times(s)	335.83	3716.0	2274.1	457.25	3618.8	1002.2	188.09	2396.8	41073	1643.3
case 3	Gaussian +impulse	MPSNR	28.91	37.32	39.52	23.18	23.39	33.16	23.44	40.09	39.35	**40.55**
		MSSIM	0.5131	0.7889	0.8627	0.4519	0.4657	0.6622	0.4749	**0.8978**	0.8163	0.8901
		ERGAS	1181.6	177.42	219.75	2158.0	2168.8	474.90	2175.1	146.87	1904.8	**142.45**
		Times(s)	328.96	3808.7	2234.6	443.50	3435.7	1105.1	186.44	2317.9	40925	1633.7
case 4	Gaussian +impulse+ deadlines	MPSNR	28.93	37.30	39.42	23.24	23.45	33.09	23.49	39.91	39.45	**40.38**
		MSSIM	0.5131	0.7905	0.8619	0.4524	0.4660	0.6615	0.4750	**0.8972**	0.8179	0.8893
		ERGAS	1181.7	176.97	220.61	2158.2	2168.4	474.93	2174.1	147.51	1864.0	**143.40**
		Times(s)	338.97	3811.4	2292.9	422.52	3361.6	1099.3	188.13	2325.9	42045	1621.6

TABLE III
QUANTITATIVE EVALUATION ON A SUBREGION OF WDC DATASET IN DIFFERENT CASES

Noise case	Level	Evaluation index	LRMR	LRTV	LRTDTV	NGmeet	GLF	SSSB	TSLRLN	LRAWGTV	FGSLR	GLRSSDA
case 1	G=0.025 P=0.05	MPSNR	39.71	41.13	38.75	32.28	32.16	38.65	32.22	41.21	**42.51**	39.44
		MSSIM	0.9820	0.9835	0.9783	0.9398	0.9369	0.9805	0.9385	0.9880	**0.9909**	0.9840
		ERGAS	43.49	51.54	47.66	103.85	104.38	50.94	104.77	41.13	**32.97**	48.11
		Times(s)	54.73	197.40	240.05	66.67	526.98	24.44	35.18	246.70	662.10	236.39
case 2	G=0.1 P=0.2	MPSNR	30.32	31.81	32.14	21.49	21.42	32.78	21.53	34.11	**35.17**	34.20
		MSSIM	0.8851	0.9000	0.9042	0.7099	0.7157	0.9279	0.7346	0.9432	**0.9526**	0.9439
		ERGAS	126.80	165.39	100.91	383.63	378.88	95.06	378.70	82.15	**71.93**	80.96
		Times(s)	55.03	203.64	232.46	72.24	512.61	80.48	35.55	244.10	2535.70	256.88
case 3	Gaussian +impulse	MPSNR	30.70	33.04	32.60	25.04	25.46	32.93	25.49	34.26	34.08	**34.86**
		MSSIM	0.8898	0.9121	0.9113	0.8102	0.8215	0.9327	0.8297	**0.9547**	0.8987	0.9528
		ERGAS	126.58	166.69	97.14	275.33	268.04	96.15	269.05	82.41	241.60	**75.44**
		Times(s)	52.19	197.84	233.35	67.90	549.65	140.57	36.37	239.80	7155.9	249.26
case 4	Gaussian +impulse+ deadlines	MPSNR	30.56	32.95	32.53	25.04	25.39	32.94	25.48	34.19	33.97	**34.73**
		MSSIM	0.8886	0.9112	0.9108	0.8087	0.8180	0.9327	0.8277	0.9545	0.8996	**0.9523**
		ERGAS	127.65	183.73	97.86	275.09	268.52	94.88	268.90	82.72	234.20	**76.38**
		Times(s)	53.86	196.09	240.88	68.78	524.63	139.90	35.86	247.26	7183.7	247.42

due to the losing of the underlying structure and detail information of HSI when using a 2-D matrix to represent the 3-D HSI dataset with large-scale. The nonlocal-based or the superpixel segmentation-based method can overcome this problem. Thus, the NGmeet, SSSB, and TSLRLN can remove most of the noise, but they loss their neighbors information, resulting in poor denoising performance in some patches. It can be observed that the denoised by NGmeet suffers from

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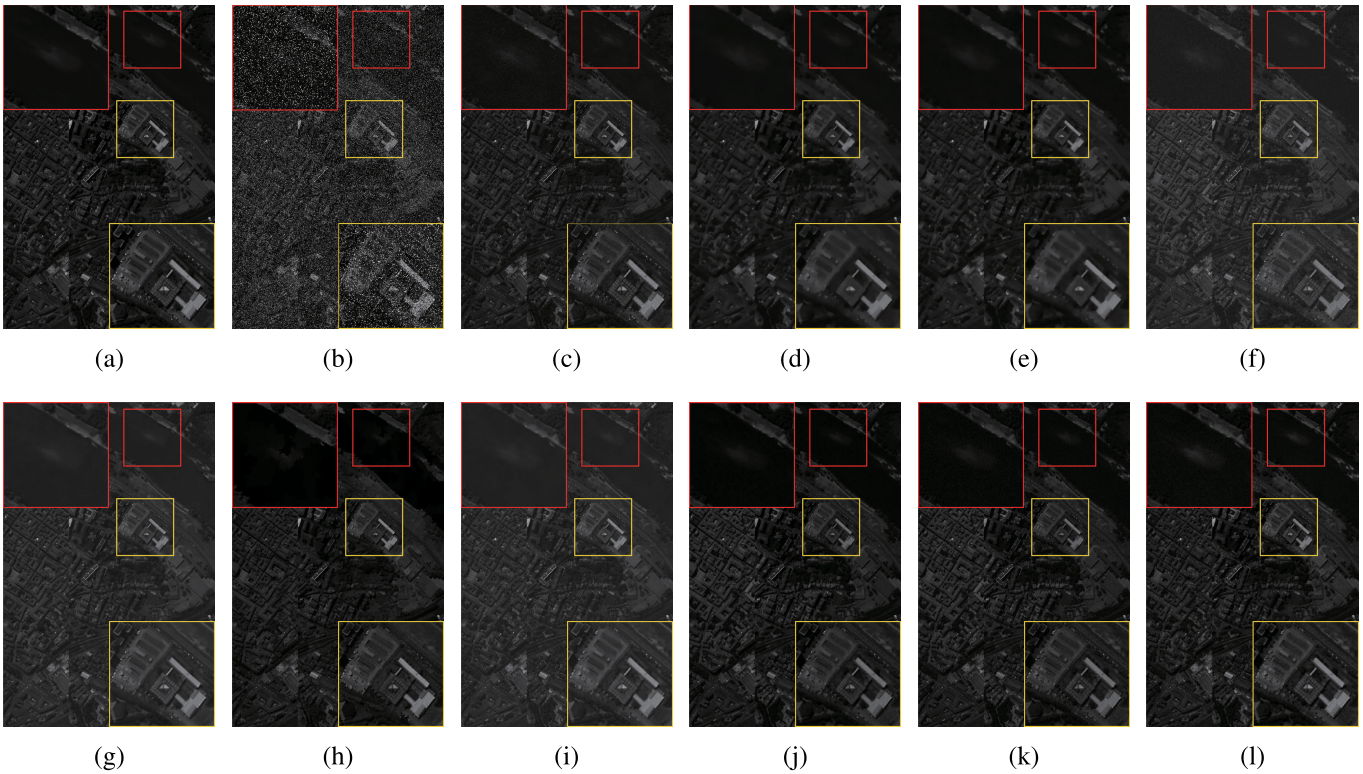


Fig. 6. Restoration results by all the compared methods of case 4 in the Pavia Center dataset. (a) Original band 8, (b) synthesis noise band, (c) LRMR, (LRTV, (e) LrTDTV, (f) NGmeet, (g) GLF, (h) SSSB, (i) TSLRLN, (j) LRAWGTV, (k) FGSLR, and (l) GLRSSDA.

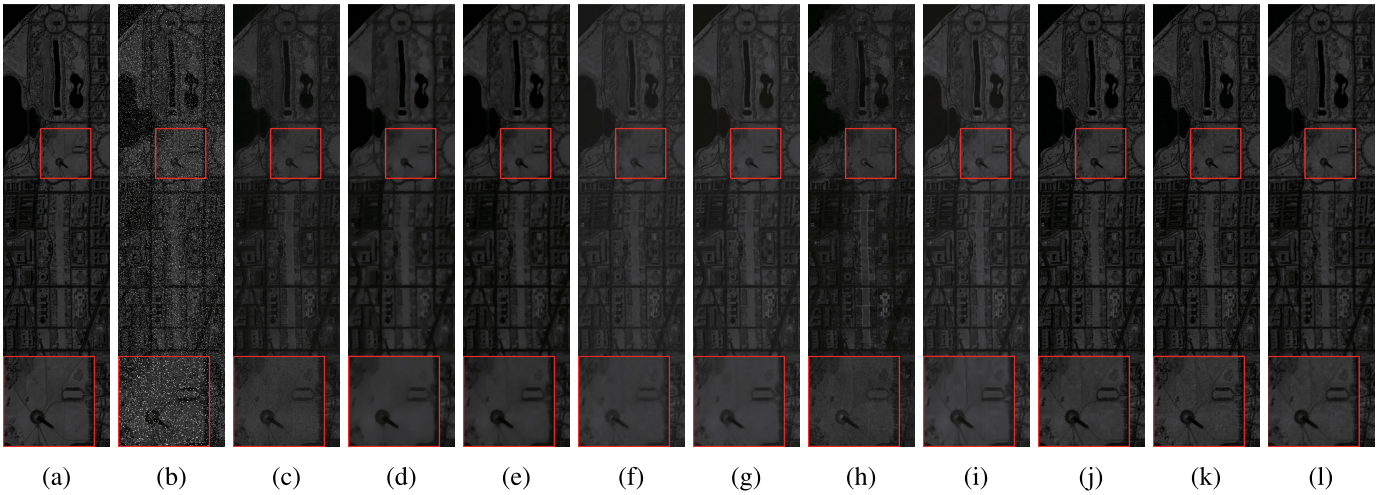


Fig. 7. Restoration results by all the compared methods of case 4 in the WDC dataset. (a) Original band 81, (b) synthesis noise band, (c) LRMR, (d) LRTV, (e) LrTDTV, (f) NGmeet, (g) GLF, (h) SSSB, (i) TSLRLN, (j) LRAWGTV, (k) FGSLR, and (l) GLRSSDA.

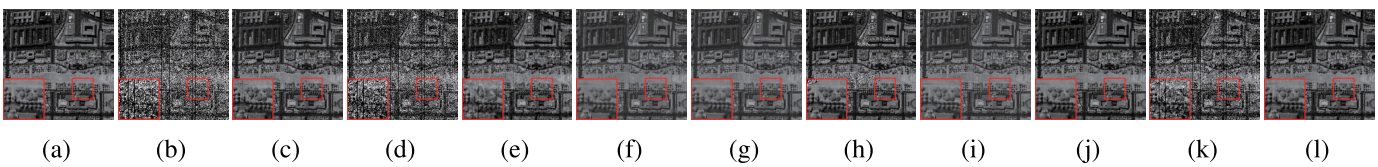
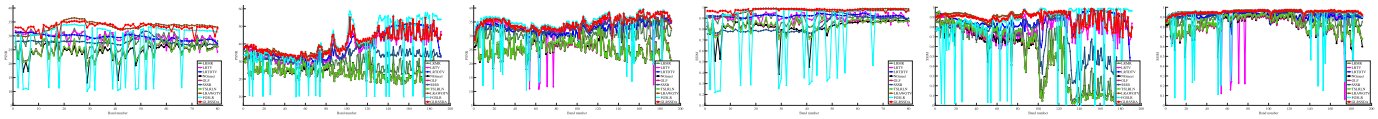


Fig. 8. Restoration results by all the compared methods of case 4 in a subregion of WDC dataset. (a) Original band 64, (b) synthesis noise band, (c) LRMR, (d) LRTV, (e) LrTDTV, (f) NGmeet, (g) GLF, (h) SSSB, (i) TSLRLN, (j) LRAWGTV, (k) FGSLR, and (l) GLRSSDA.

over-smoothing and the denoised HSI by TSLRLN contains noise in some patches. The superpixel segmentation-based method SSSB performs well, but it is the same with the other methods that it loses the detailed information of the HSI, such as the texture information (small and long island) in the top of the enlarged box. The proposed GLRSSDA provides the best performance, it remove the noise, while preserving the detailed information.

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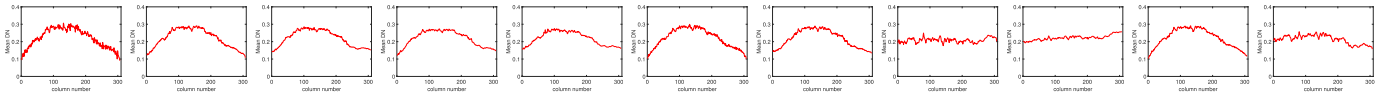
(a) (b) (c) (d) (e) (f)

Fig. 9. (a) PSNR values of each band with the Pavia Center dataset in Case 4. (b) PSNR values of each band with the WDC dataset in Case 4. (c) PSNR values of each band with a subregion of WDC dataset in Case 4. (d) SSIM values of each band with the Pavia Center dataset Case 4. (e) SSIM values of each band with the WDC dataset in Case 4. (f) SSIM values of each band with a subregion of WDC dataset in Case 4.



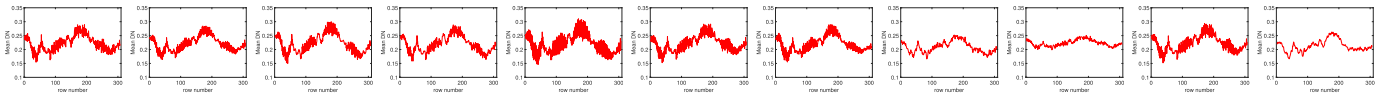
(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k) (l)

Fig. 10. Restoration results of the HYDICE Urban real dataset. (a) False color image (R: 80, G: 104, B: 150), (b) original band 104, (c) LRMR, (d) LRTV, (e) LRTDTV, (f) NGmeet, (g) GLF, (h) SSSB, (i) TSLRLN, (j) LRAWGTV, (k) FGSLR, and (l) GLRSSDA.



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k)

Fig. 11. Vertical mean profiles of band 104 in HYDICE Urban real dataset. (a) Original, (b) LRMR, (c) LRTV, (d) LRTDTV, (e) NGmeet, (f) GLF, (g) SSSB, (h) TSLRLN, (i) LRAWGTV, (j) FGSLR, and (k) GLRSSDA.



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k)

Fig. 12. Horizontal mean profiles of band 104 in HYDICE Urban real dataset. (a) Original, (b) LRMR, (c) LRTV, (d) LRTDTV, (e) NGmeet, (f) GLF, (g) SSSB, (h) TSLRLN, (i) LRAWGTV, (j) FGSLR, and (k) GLRSSDA.

C. Discussion

In the proposed GLRSSDA method, the denoising performance is influenced by the values of various of parameters. In the preprocessing step, the median filter is suggested to use a 3×3 window. In the SLIC algorithm, the regularizer of the SLIC is set to 0.05 for all the experiments of the proposed method. For the size of the superpixel, it is related to the parameter "region size" of the SLIC algorithm [53], [54], where the region size is selected empirically according to the size of the dataset. In the proposed method, the region size is 15 for the subregion of WDC dataset, 20 for the Urban dataset, and 30 for the rest datasets. In the modified KNN algorithm, the K is set to 3 for all the experiments, the larger K, the more neighbor information is considered, the larger computing complexity is. Thus, K can be larger, if there is no requirement for computing complexity. Note that K is not the number of neighbors, due to the symmetrization operation in modified KNN algorithm. In optimization problem of the subgraphs, the parameter α is in the range of $[0.01, 1]$, r is set to 3. The parameter β in all experiments is selected according to the size of the subgraph, where we simply set $\beta = 0.25$ for the dataset of this article. Next, we explain the reasons for choosing these parameter settings. All the experiments are conducted on the simulated datasets under Case 3.

1) Analysis of α and β: In GLRSSDA, parameter α is used to restrict the proportion of GLR in the model. The larger α is, the stronger power of the GLR regularization is. However, when α is too large, the denoised HSI will be over-smoothing. Thus, a suitable α can improve the denoising performance. To determine the optimal α, we use a cross-validation approach. In specific, we fix parameters other than α, and compare the impact of different α on MPSNR and MSSIM. The result is shown in Fig. 16. Combining the denoising results in simulated datasets with the results in Fig. 16, it can be seen that α is closely related to the PWS property of the dataset. When the dataset has more details and is less smoothness, a smaller α will be more suitable, and vice versa. For simplicity, we can set $\alpha = 0.1$ for the visually nonsmooth dataset, and set $\alpha = 0.5$ for the visually smooth dataset. β is the parameter used to restrict the sparsity of the sparse noise, which is related to the size of dataset [21]. Owing to the superpixel segmentation and modified KNN algorithm, the number of pixels belonging to each subgraph is similar. Thus, the same β can be selected for different datasets.

TABLE IV
ABLATION EXPERIMENT RESULTS

Evaluation index	AE1	AE2	GLRSSDA
MPSNR	34.12	21.14	**34.86**
MSSIM	0.9445	0.6460	**0.9528**
ERGAS	84.76	377.92	**75.44**

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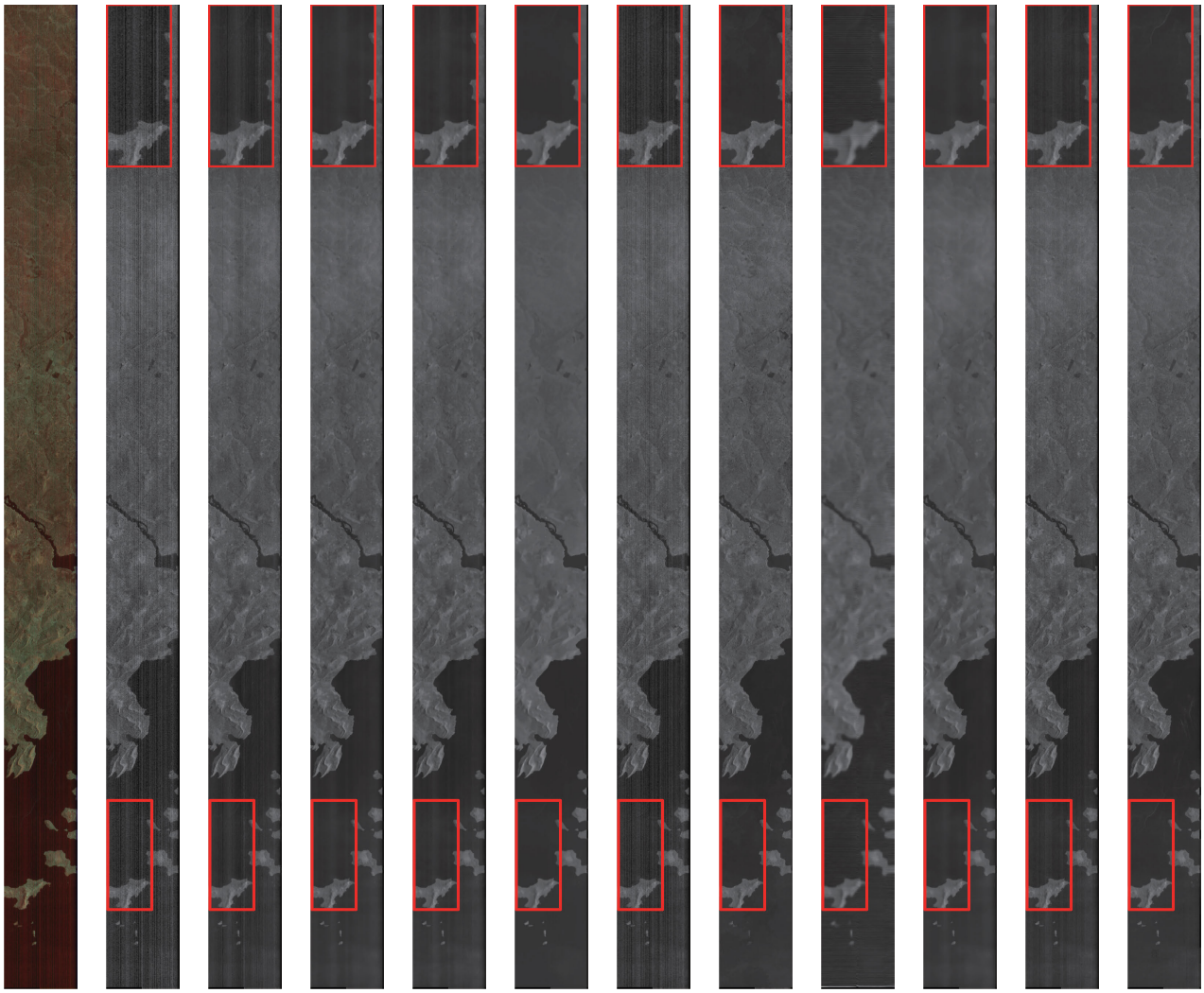


Fig. 13. Restoration results by all the compared methods in EO-1 Hyperion real dataset. (a) False color image (R: 119, G: 50, B: 91), (b) original band 119, (c) LRMR, (d) LRTV, (e) LRTDTV, (f) NGmeet, (g) GLF, (h) SSSB, (i) TSLRLN, (j) LRAWGTV, (k) FGSLR, and (l) GLRSSDA.

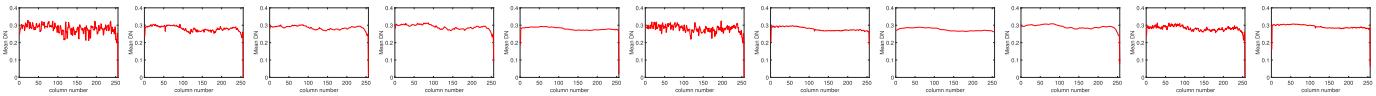


Fig. 14. Vertical mean profiles of band 119 in EO-1 Hyperion real dataset. (a) Original, (b) LRMR, (c) LRTV, (d) LRTDTV, (e) NGmeet, (f) GLF, (g) SSSB, (h) TSLRLN, (i) LRAWGTV, (j) FGSLR, and (k) GLRSSDA.

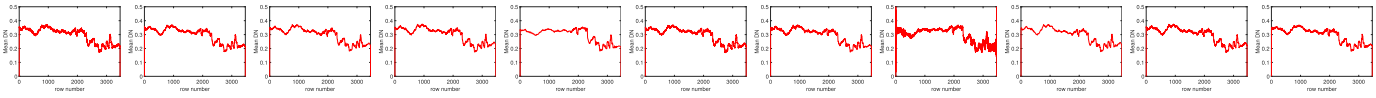
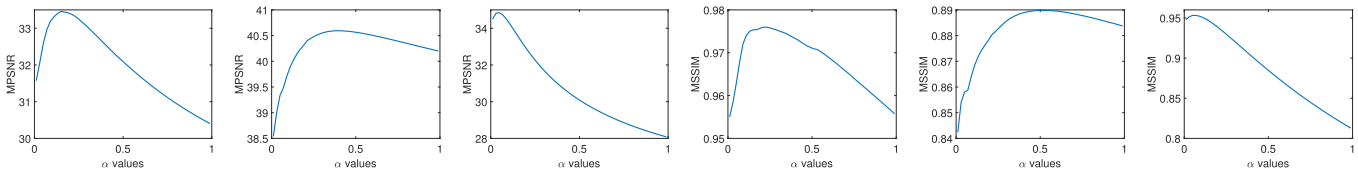


Fig. 15. Horizontal mean profiles of band 119 in EO-1 Hyperion real dataset. (a) Original, (b) LRMR, (c) LRTV, (d) LRTDTV, (e) NGmeet, (f) GLF, (g) SSSB, (h) TSLRLN, (i) LRAWGTV, (j) FGSLR, and (k) GLRSSDA.

To determine the optimal β, we also use a cross-validation approach. In specific, we fix parameters other than β, and compare the impact of different β on MPSNR and MSSIM. The result is shown in Fig. 17. From Fig. 17, we can set $\beta = 0.25$.

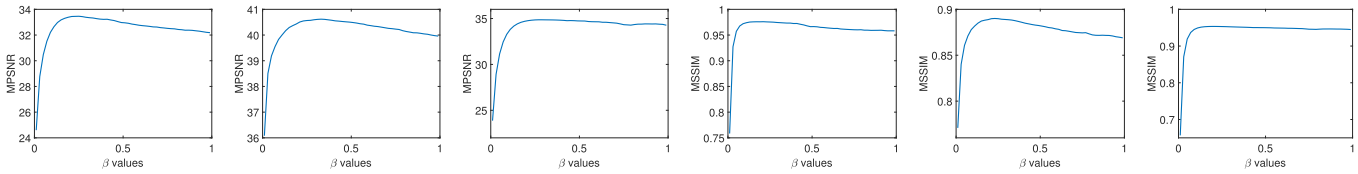
2) Analysis of r: In GLRSSDA, parameter r is used to pilot the number of nonzero singular values in SVD when we resolve the nuclear norm. Thus, it is related to the size of dataset [21]. Owing to the superpixel segmentation and modified KNN algorithm, the number of pixels belonging to

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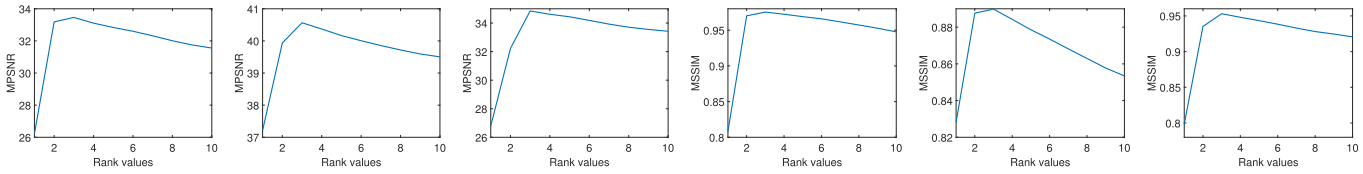
(a) (b) (c) (d) (e) (f)

Fig. 16. Sensitivity analysis of parameter α (α from 0.01 to 1, step size is 0.02). (a) Change in the MPSNR value with the Pavia Center dataset. (b) Change in the MPSNR value with the WDC dataset. (c) Change in the MPSNR value with a subregion of WDC dataset. (d) Change in the MSSIM value with the Pavia Center dataset. (e) Change in the MSSIM value with the WDC dataset. (f) Change in the MSSIM value with a subregion of WDC dataset.



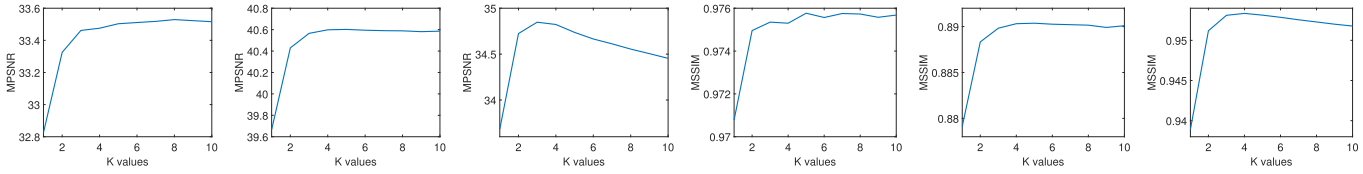
(a) (b) (c) (d) (e) (f)

Fig. 17. Sensitivity analysis of parameter β (β from 0.01 to 1, step size is 0.02). (a) Change in the MPSNR value with the Pavia Center dataset. (b) Change in the MPSNR value with the WDC dataset. (c) Change in the MPSNR value with a subregion of WDC dataset. (d) Change in the MSSIM value with the Pavia Center dataset. (e) Change in the MSSIM value with the WDC dataset. (f) Change in the MSSIM value with a subregion of WDC dataset.



(a) (b) (c) (d) (e) (f)

Fig. 18. Sensitivity analysis of parameter r (r from 1 to 10). (a) Change in the MPSNR value with the Pavia Center dataset. (b) Change in the MPSNR value with the WDC dataset. (c) Change in the MPSNR value with a subregion of WDC dataset. (d) Change in the MSSIM value with the Pavia Center dataset. (e) Change in the MSSIM value with the WDC dataset. (f) Change in the MSSIM value with a subregion of WDC dataset.



(a) (b) (c) (d) (e) (f)

Fig. 19. Sensitivity analysis of parameter K (K from 1 to 10). (a) Change in the MPSNR value with the Pavia Center dataset. (b) Change in the MPSNR value with the WDC dataset. (c) Change in the MPSNR value with a subregion of WDC dataset. (d) Change in the MSSIM value with the Pavia Center dataset. (e) Change in the MSSIM value with the WDC dataset. (f) Change in the MSSIM value with a subregion of WDC dataset.

TABLE V

RUN TIMES (IN S) OF THE DIFFERENT METHODS IN THE REAL DATASET

Method	LRMR	LRTV	LRTDTV	NGmeet	GLF	SSSB	TSLRLN	LRAWGTV	FGSLR	GLRSSDA
Urban	99.99	382.27	440.86	111.83	1641.9	452.42	87.71	462.29	5886.7	402.63
EO-1	970.99	9248.0	9259.4	919.72	19525	2884.3	1134.8	9214.4	12268	5509.9

each subgraph is similar. Thus, the same r can be selected for different datasets. To determine the optimal r, we fix parameters other than r, and compare the impact of different r on MPSNR and MSSIM. The result is shown in Fig. 18. From Fig. 18, we can set $r = 3$.

3) Analysis of K: Parameter K is used to control the size of subgraphs. The larger K is, the larger size of the subgraphs is. However, the larger K will lead to data redundancy. Thus, a suitable K can not only reduce the computing cost but also improve the performance. To determine the optimal K, we fix parameters other than K, and compare the impact of different K on MPSNR and MSSIM. The result is shown in Fig. 19. From Fig. 19, it can be seen that when $K = 3$, the algorithm can perform well while costing less.

4) Ablation Experiment: We conduct two ablation experiments on the subregion of WDC dataset under case 3.

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The first experiment is that we replace superpixels with square patches (AE1). The second experiment is that we remove the two-layer graph architecture, i.e., the denoising problem is solved by one superpixel followed by another, which cannot effectively consider the neighbor information (AE2). The experiment results are shown in Table IV.

From Table IV, it can be seen that superpixel segmentation can improve denoising performance owing to the preservation of boundary information. Besides, the two-layer graph architecture with a distributed algorithm can effectively characterize the underlying structure of HSI, in which the distributed algorithm with neighbor information exchange can achieve better denoising performance while costing less-computing resources.

5) Time Cost: Table V presents the time cost of the different methods for the two real datasets. Combine Tables I–III with Table V, although the proposed method spends more time than some methods with simple model, the nonlocal-based methods, or the superpixel segmentation-based methods, the proposed GLRSSDA can achieve more excellent denoising performance. And, the GLRSSDA runs faster than the methods that achieve the similar denoising performance. Besides, the GLRSSDA is designed in distributed framework, it can achieve faster parallel computing while there are more cooperative physical computing units.

VI. Conclusion

In this article, we proposed a distributed method for HSI mixed-noise removal, which extended the application of GSP theory to HSI restoration domain. A two-layer architecture graph was proposed to characterize the underlying structure of the HSI. First, a superpixel segmentation was applied to segment the HSI into a series of independent (nonoverlapping) and irregular superpixels. Then, a skeleton graph was constructed by a modified KNN algorithm, which captured the similarity between superpixels. Besides, the detailed graph consisting of a series of subgraphs were constructed to model the similarities between pixels. Based on the two-layer architecture graph, the HSI restoration problem was formulated as a series of optimization problems each of which resided on a subgraph and incorporated GLR into an LR-based model. A distributed algorithm was proposed to solve the problems, which can overcome the rapidly increase of computational complexity of the restoration of HSI, particularly when the HSI is of large scale. The proposed method was compared with the state-of-the-art restoration methods in many experiments, demonstrating that the proposed GLRSSDA almost performed the best in terms of both visual and quantitative estimation in both small and large HSI dataset.

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